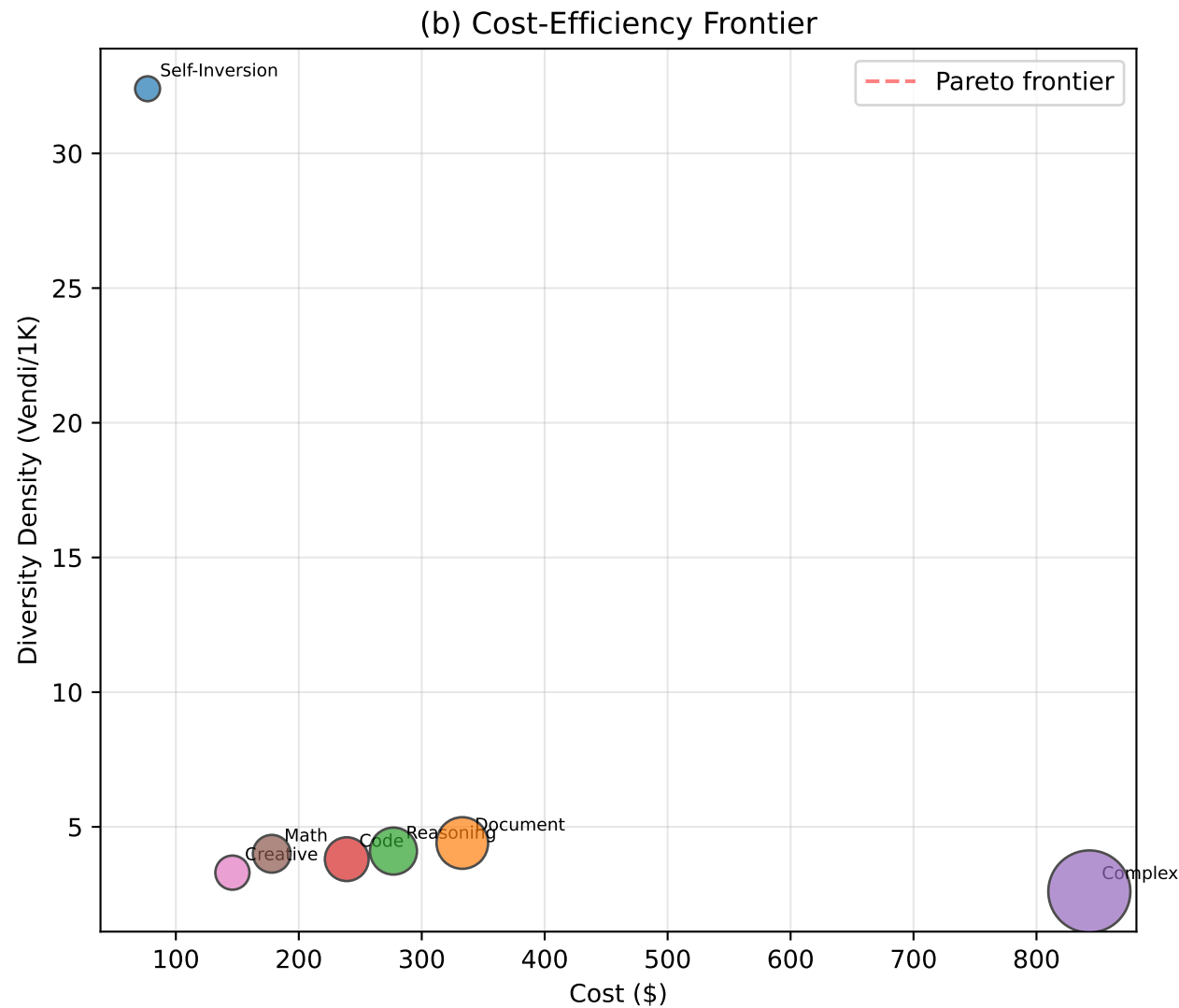
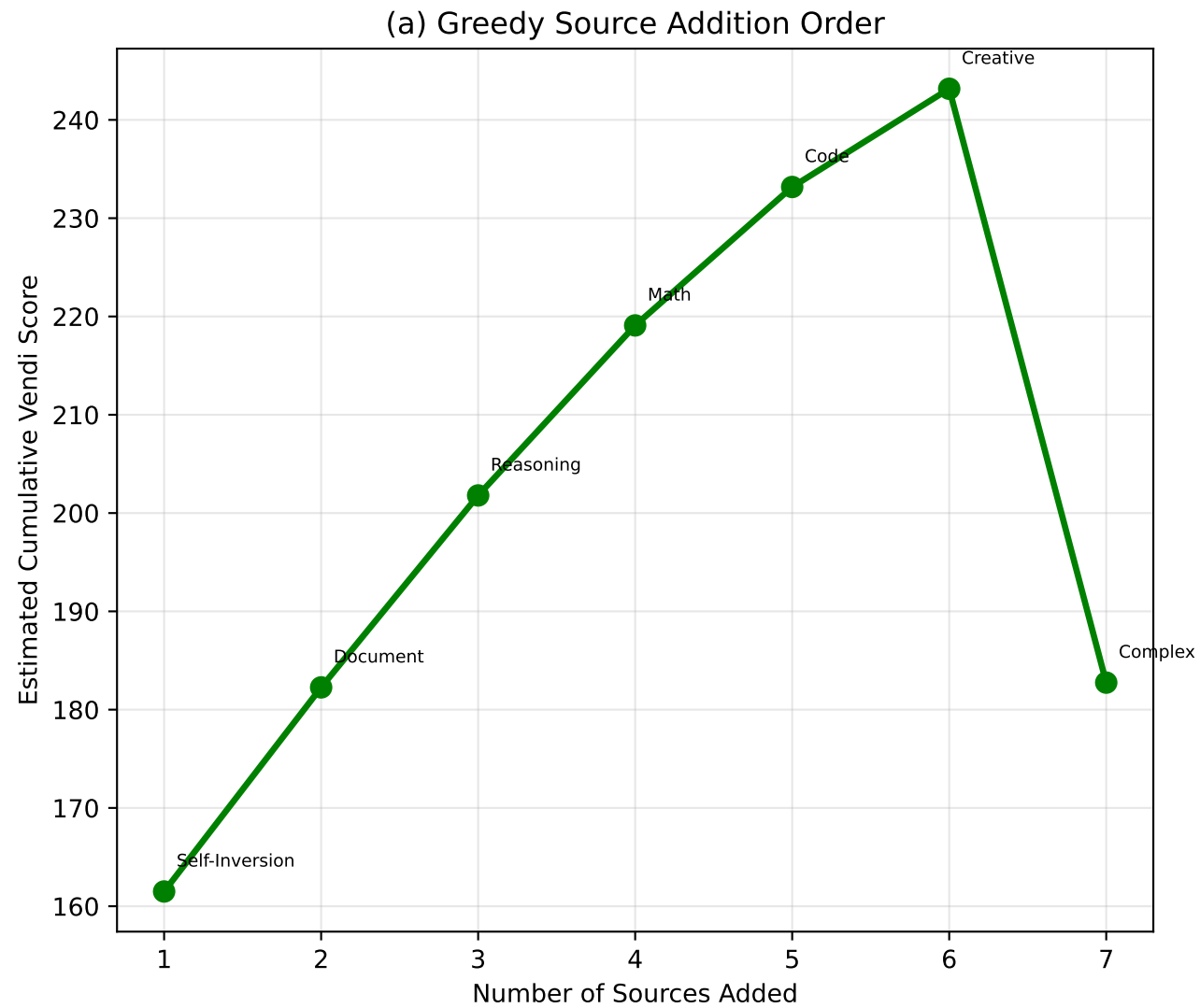
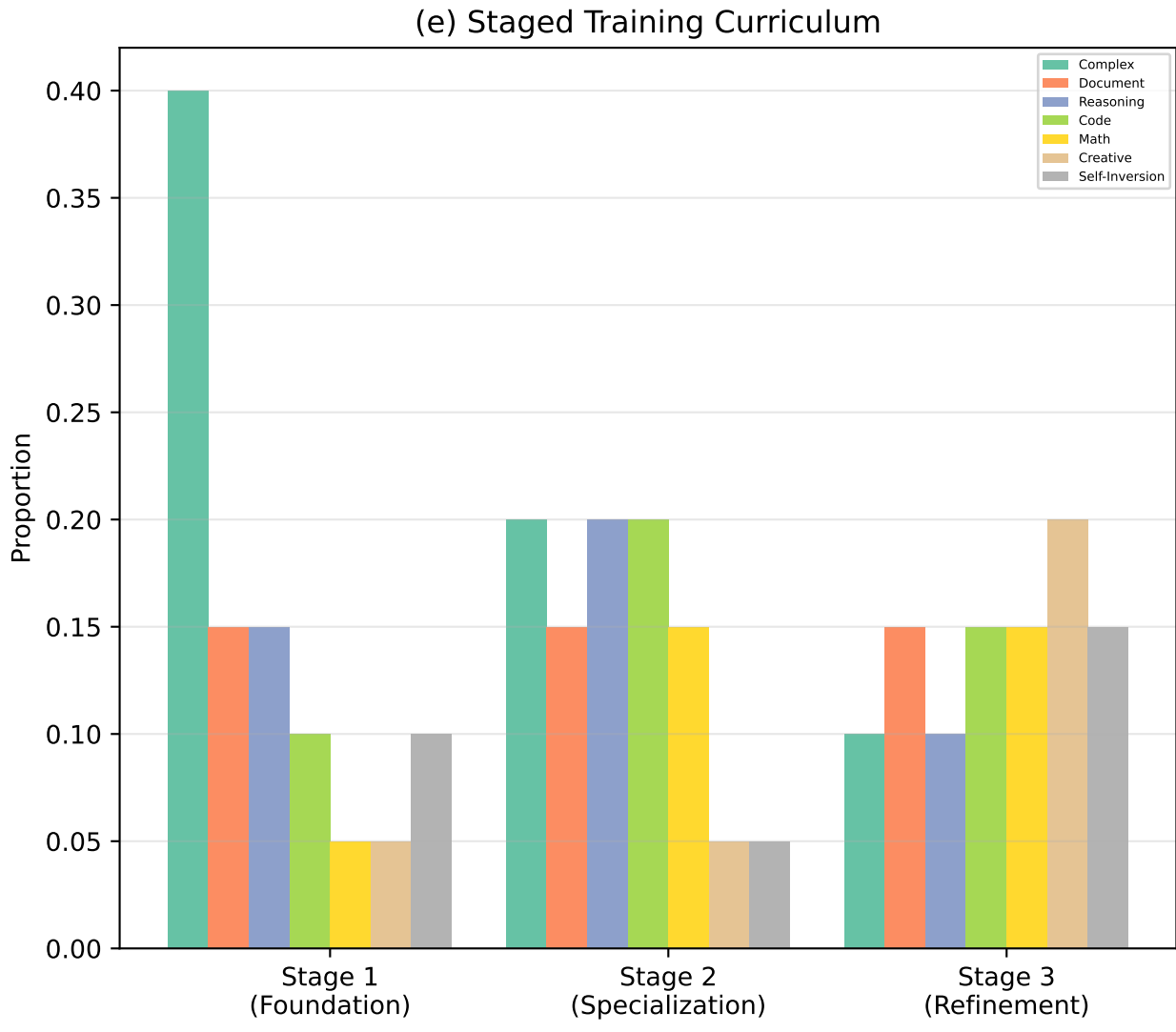
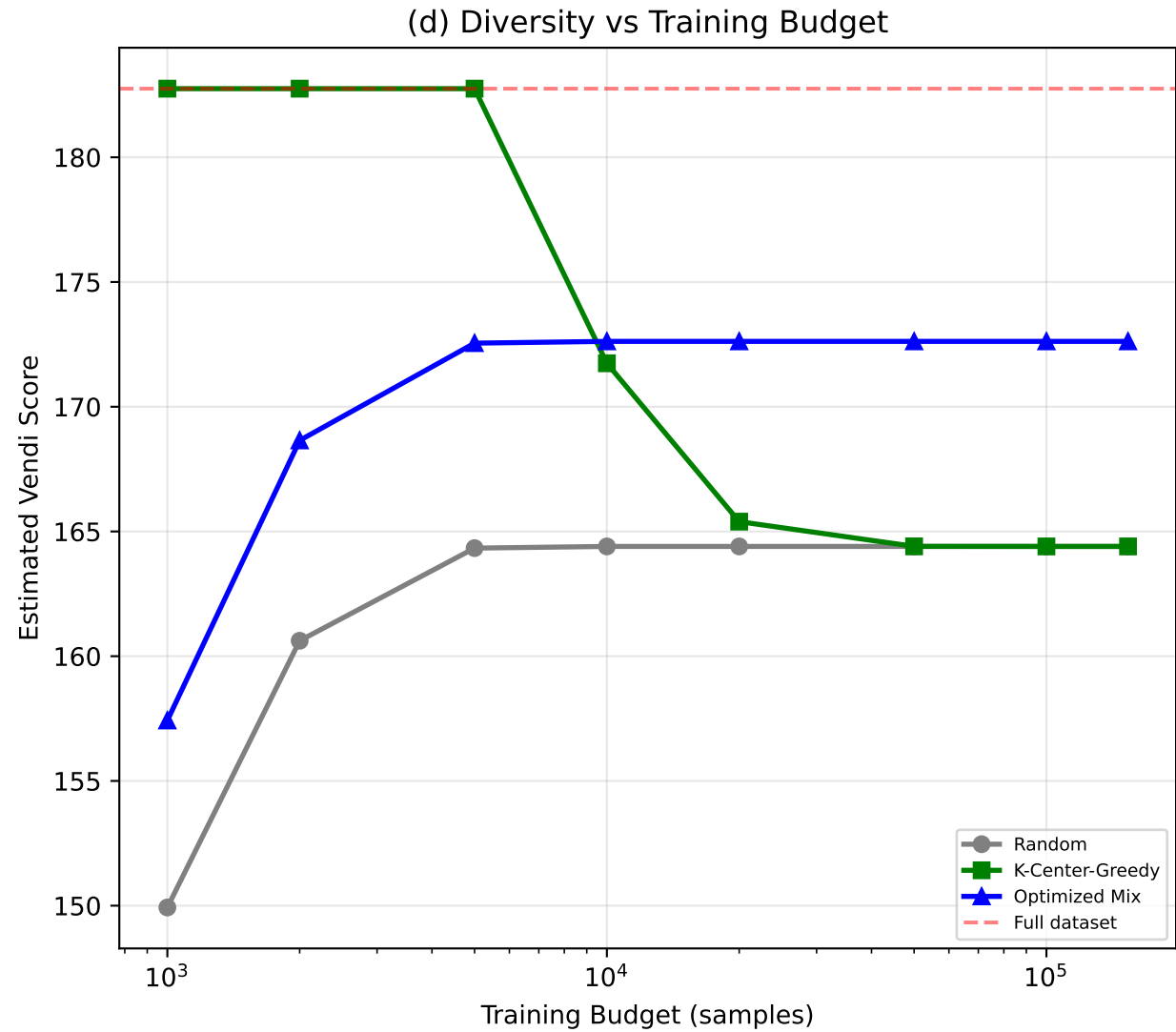


Fig 130: Optimal Training Curriculum & Budget Allocation



(c) Budget-Based Recommendations

Budget	Est. Samples	Recommended Mix	Est. Vendi
\$100 budget	5K samples	Self-Inv (all) + Doc QA (part)	VS $\approx$ 170
\$500 budget	30K samples	Self-Inv + Doc + Reasoning + Code	VS $\approx$ 178
\$1,000 budget	65K samples	All 7 + emphasis on Self-Inv	VS $\approx$ 181
\$2,245 (full)	153K samples	All 8 pipelines	VS = 182.75
\$500 (opt)	30K optimized	Upweight Self-Inv to 17.4%	VS $\approx$ 180



- (f) Top 10 Recommendations
1. Start with Self-Inversion for maximum diversity per dollar
 2. Add Document QA next for linguistic quality & grounding
 3. Use K-Center-Greedy for small budgets ($N < 5K$)
 4. Random sampling suffices for $N > 5K$
 5. Upweight Self-Inv from 3.2% $\rightarrow$ 17.4% for diversity
 6. Downweight Complex QA from 35.4% $\rightarrow$ 13.7%
 7. Include Multi-Turn for conversation capability
 8. Total cost: 2,245for153Ksamples(15.58/1K)
 9. 90% diversity achievable with just 5K samples (K-Center)
 10. 0 benchmark contamination – safe for evaluation